How to Install Images

Introduction

This guide assumes the use of the provided Multiboot_Admin environment to install additional images, although the same approach is used to install images using an external Linux os. This os should auto-login with two console screens but in a gui environment. For info **user=pi, password=pi**

1) Install or update the Multiboot Toolchain

- If using the Multiboot_Admin os then this toolchain comes pre-installed at /home/pi/Pi_Multiboot. However you should make sure that it is up to date by running

```
cd ~
./update_imager.sh
```

- If using an externel os you can fetch and then run the same script

```
cd ~
wget https://raw.githubusercontent.com/AndyDeSheffield/Pi_Multiboot/refs/heads/main/tools/update_imager.sh
sudo chmod +x update_imager.sh
./update_imager.sh
```

2) Mount the images partition which will receive your new os

- When using the Multiboot_Admin os this is most likely your current images partition. In this case it can be mounted using the script in the Pi_Multiboot/tools directory

```
Pi_Multiboot/tools/mountdrives.sh
```

your target partition will be /mnt/realimages

- Otherwise you will likely be installing to an external usb key partition (example /dev/sdb3) so

```
mkdir -p ~/realimages
sudo mount /dev/sdb3 ~/realimages
```

your target partition is of course ~/realimages

3) Run the image creation tool

- First check the help
- Note that after each action the imaging tool gives the option to continue or abort (which backs out previous steps)

```
cd ~/Pi_Multiboot/create_images
./install_image.sh -h

Usage: $0 -r=<root_mount> -n=<name_of_image> -m=<pi_model> -s=<size> [-e] [-h]
  -r=<root_mount>    Root of mounted image disk (e.g. -r=/mnt/realimages)
  -n=<name>          Name of image (letters, numbers, '-', '_')
  -m=<pi_model>      Pi model (e.g. pi3b,pi3b+,pi4, pi5)
  -s=<size>          Image size (e.g. 10G)
  -c                 Apply custom files to the image directory (will not hurt if there are no custom files)
  -e                 Expand root filesystem (only if exactly 2 partitions exist)
```

```
  -h                  Show this help
  Note that the script needs to be run with sudo and relies on the device tree compiler (DTC) being available
```

-
- Run the tool as sudo with the appropriate parameters

```
sudo ./install_image.sh -r=/mnt/realimages -n=My_New_OS -m=pi4 -s=10G -e -c
```

1. Step 1 creates the blank .img file on the root Mount

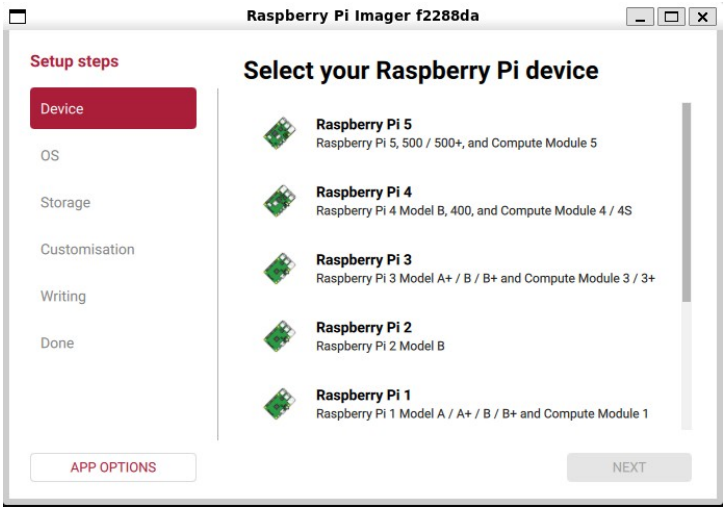
```
Step 1: Creating target directory and blank image...
Created image: /mnt/realimages/My_New_OS_pi4/My_New_OS_pi4.img (size 10G)

Step 1 completed. Continue (c) or Abort (a)? [c/a]:
```

1. Step 2 runs a modified version of the raspi-imager that accepts loop mounts as targets

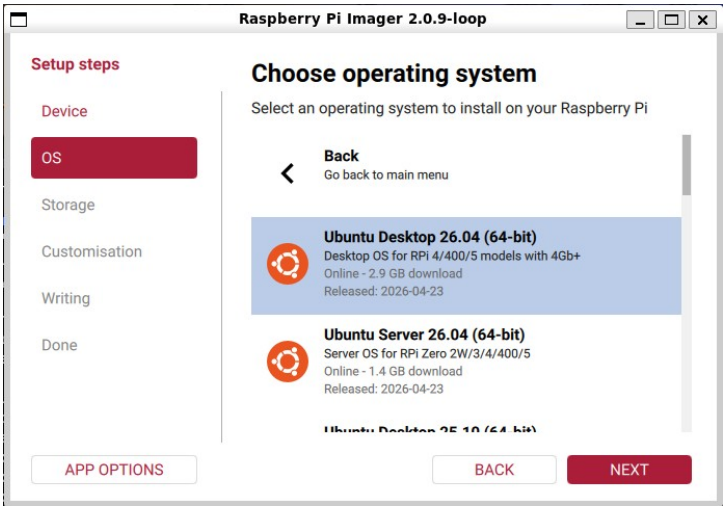
```
Continuing...
Step 2: Loop-mounting image and running Raspberry Pi Imager...
Loop device: /dev/loop2
Launching Raspberry Pi Imager...
Refreshing loop partitions...
Reattached loop device: /dev/loop2
```

Follow the usual imaging process selecting the loop device attached to your image as the target

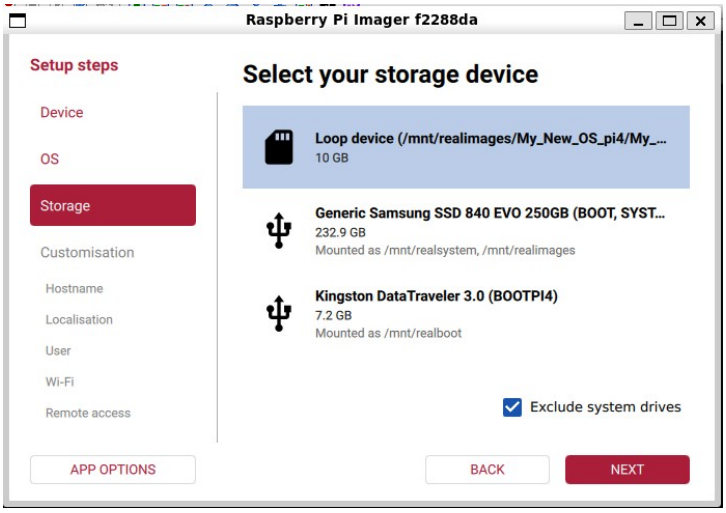
- Select Pi model



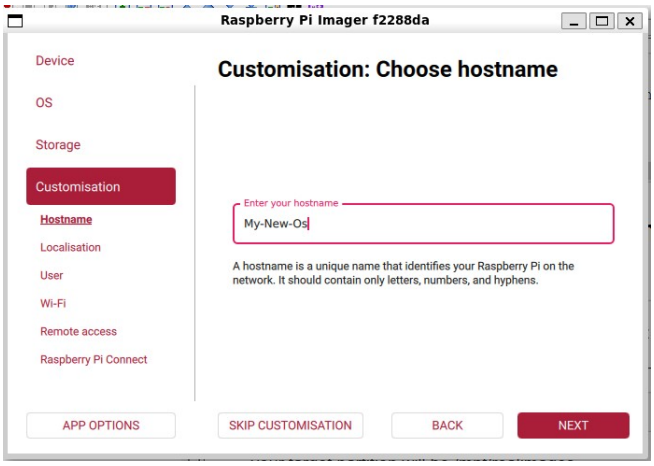
- Select Operating System

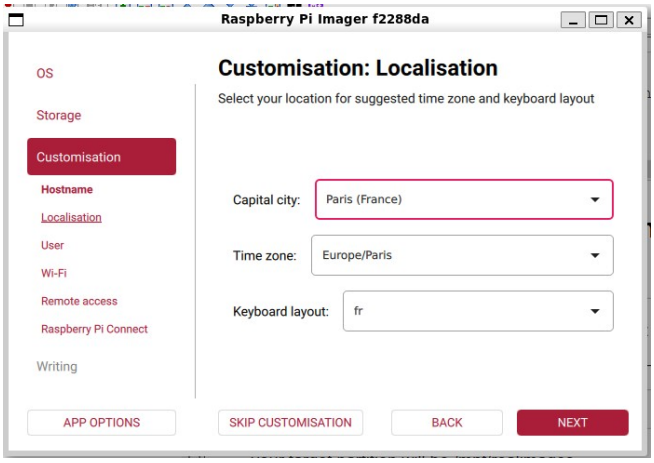


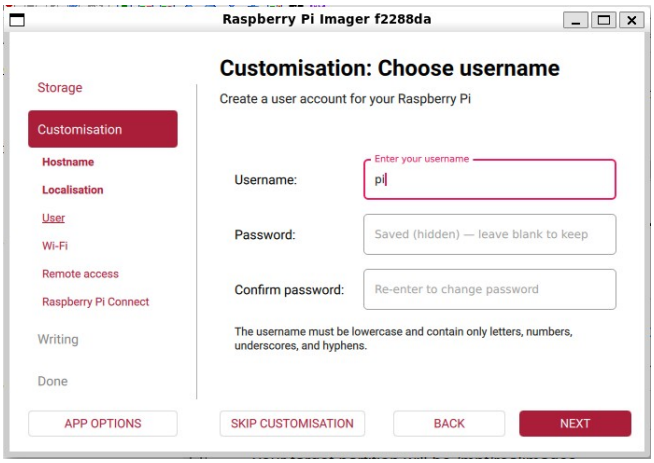
Select Target loop --- **Be careful NOT to select the SSD or Hard drive itself!**

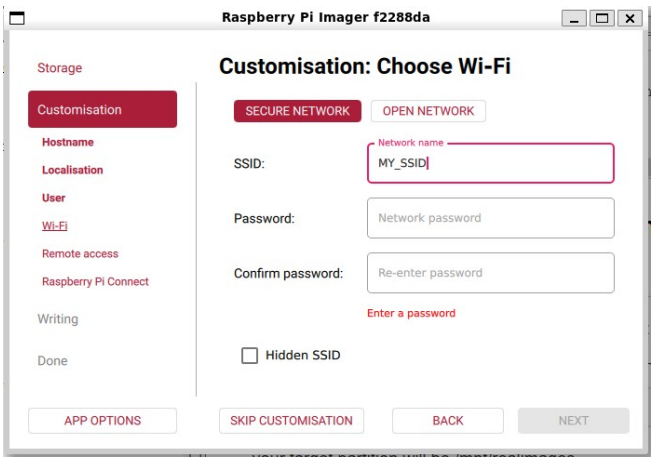


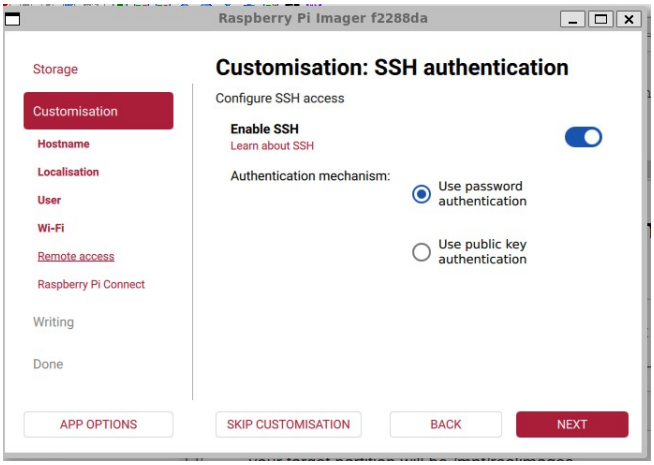
- Configure if desired



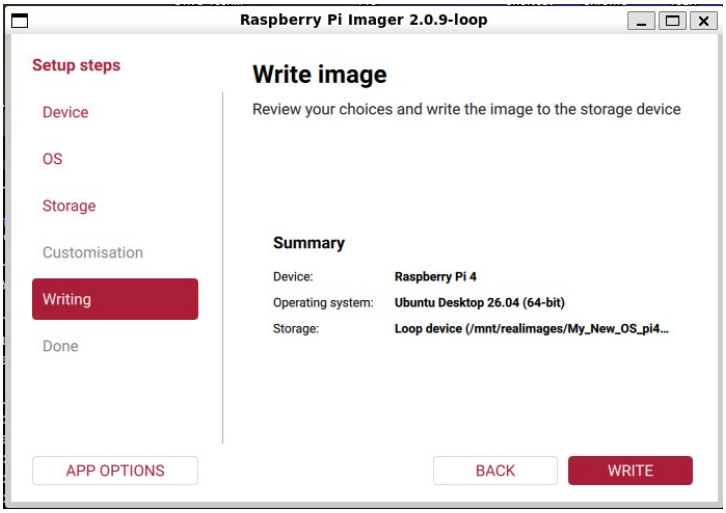




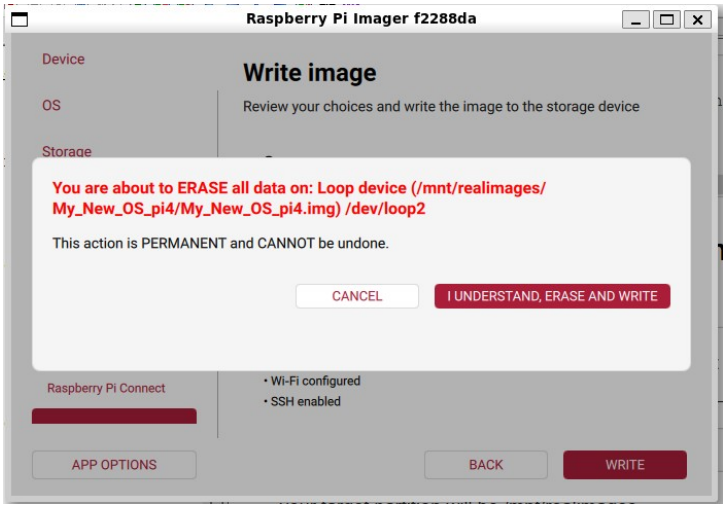




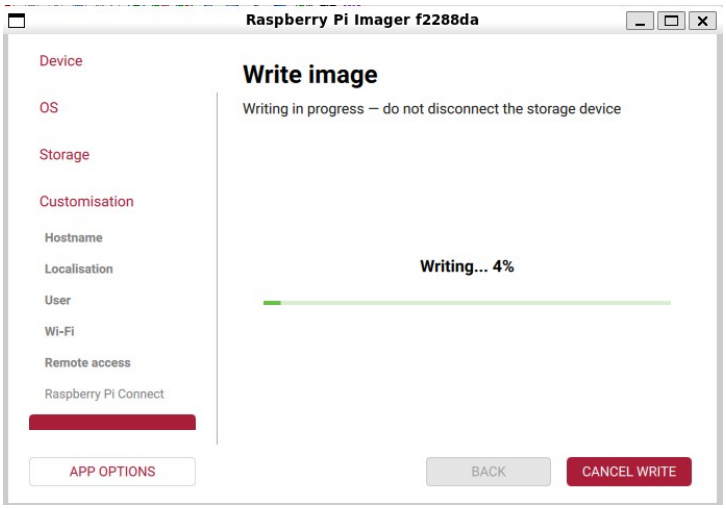
- Write Image



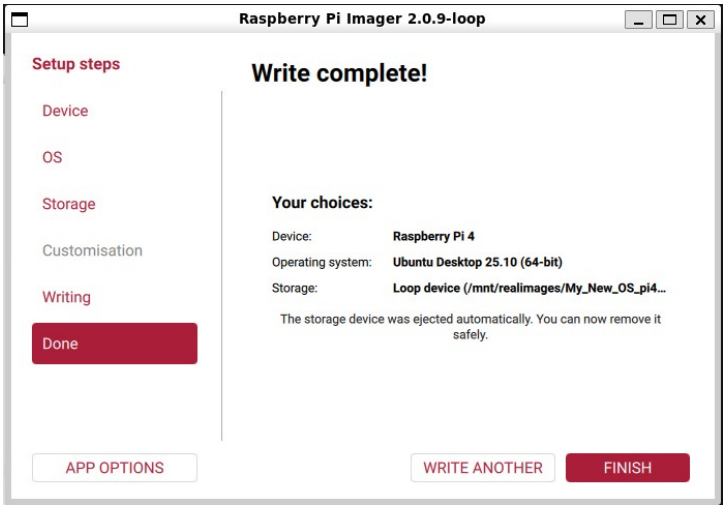
- Confirm Write



- Writing



- Finish




```
Step 2 completed. Continue (c) or Abort (a)? [c/a]:
```

1. Step 3 creates a dtb file with all the overlays and properties specified in the config.txt of the target boot partition merged in.
    - It copies that dtb, along with the target kernel and any initramfs into a directory /target_name on the root mount.
    - A suitable Grub entry file is also created
    - The kernel cmdline used in the official image is extracted for reference. Use the "extras" variable in the grub config to add any desired values.
    - The config.txt from the original boot partition is extracted for reference and a file "untreated_config.txt" is created showing lines that it has not been possible to process.
    - If the -e option is selected the root partition of the new image will be expanded to fill the img file
    - If the -c option is selected then custom files for the image will be copied into the directory where the img file is hosted. Details of what will be copied are displayed and you can confirm the copy before it is made

```
Continuing...
Step 3: Extracting DTB and kernel...
Handling uefi fixes. Installing cpufix.dtbo, dmafix.dtbo and uefi_fixes.txt
Model: pi4 (index 5)
Boot directory: ./bootpart
Will output to : /mnt/realimages/My_New_OS_pi4/My_New_OS_pi4{.dtb,.kernel,.cmdline,.initrd}
Any kernel, cmdline and initrd files will be copied (-x specified.)


UNTREATED CONFIG LINES stored in  /mnt/realimages/My_New_OS_pi4/untreated_config.txt
uefi_fixes.txt found and will be processed after config.txt
COPYING FULL Config.txt ./bootpart/config.txt to /mnt/realimages/My_New_OS_pi4/config.txt
COPYING SPECIFIED KERNEL ./bootpart/current/vmlinuz to /mnt/realimages/My_New_OS_pi4/My_New_OS_pi4.kernel
COPYING INITRAMFS ./bootpart/current/initrd.img to /mnt/realimages/My_New_OS_pi4/My_New_OS_pi4.initrd
SET_BASE_PROPERTY (group 0): audio=on
SET_BASE_PROPERTY (group 1): i2c_arm=on
SET_BASE_PROPERTY (group 2): spi=on
SET_BASE_PROPERTY (group 3): watchdog=on
Warning: unknown base dtparam 'watchdog' ignored
LOAD_OVERLAY (group 4): ./bootpart/current/overlays/vc4-kms-v3d-pi4.dtbo
APPLY_OVERLAY (group 4): ./bootpart/current/overlays/vc4-kms-v3d-pi4.dtbo
LOAD_OVERLAY (group 5): ./bootpart/current/overlays/dwc2.dtbo
APPLY_OVERLAY (group 5): ./bootpart/current/overlays/dwc2.dtbo
LOAD_OVERLAY (group 6): ./bootpart/current/overlays/miniuart-bt.dtbo
APPLY_OVERLAY (group 6): ./bootpart/current/overlays/miniuart-bt.dtbo
LOAD_OVERLAY (group 7): ./bootpart/current/overlays/cpufix.dtbo
APPLY_OVERLAY (group 7): ./bootpart/current/overlays/cpufix.dtbo
LOAD_OVERLAY (group 8): ./bootpart/current/overlays/dmafix.dtbo
```

```
APPLY_OVERLAY (group 8): ./bootpart/current/overlays/dmafix.dtbo
COPYING MERGED DTB to /mnt/realimages/My_New_OS_pi4/My_New_OS_pi4.dtb
Files sucessfully created
Step 3.7: Optional expansion...
Expanding root filesystem...
Model: Loopback device (loopback)
Disk /dev/loop2: 10.7GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

Number  Start   End     Size    Type     File system  Flags
 1      1049kB  538MB   537MB   primary  fat32        boot, lba
 2      538MB   8922MB  8384MB  primary  ext4

e2fsck 1.47.2 (1-Jan-2025)
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 3A: Optimizing directories
Pass 4: Checking reference counts
Pass 5: Checking group summary information

writable: ***** FILE SYSTEM WAS MODIFIED *****
writable: 121987/512064 files (0.1% non-contiguous), 1377596/2046785 blocks
resize2fs 1.47.2 (1-Jan-2025)
Resizing the filesystem on /dev/loop2p2 to 2490112 (4k) blocks.
The filesystem on /dev/loop2p2 is now 2490112 (4k) blocks long.

Expansion complete.
[cleanup] Forcing release of ./bootpart...
Generating GRUB entry: /mnt/realimages/My_New_OS_pi4/My_New_OS_pi4_grub.cfg
GRUB entry written to: Grub_My_New_OS_pi4.cfg
Running customisation (may overwrite generated grub entry)...
 sudo ./deploy_custom_files.py /mnt/realimages/My_New_OS_pi4/My_New_OS_pi4.img
[detect] OS: ubuntu, VERSION_ID: 25
[info] OS=ubuntu  version=25  (numeric 25.0)
[info] UUIDs: {1: '0808-11DE', 2: 'c418dc3c-288d-475d-b6d2-48008dea7829'}
[match] Using files from: /home/pi/Pi_Multiboot/create_images/custom_files/ubuntu/25+

  Custom files from : /home/pi/Pi_Multiboot/create_images/custom_files/ubuntu/25+
  Will be copied to : /mnt/realimages/My_New_OS_pi4
  Prefixed with     : My_New_OS_pi4_

Continue? [Y/n]: Y
[deploy] Deployed 2 file(s) to /mnt/realimages/My_New_OS_pi4
[done] Complete.
```

4) Insert the contents of the suggested Grub file into grub.cfg

- If using the Multiboot_Admin os and you have mounted the target partitionusing mountdrives.sh you are already in a position to do this.
- Otherwise you have to insert the device that you will be using and mount it. The grub.cfg file that needs updating will be /efi/boot/grub.cfg
- Note that it may, in any case be preferable to update /mnt/realboot/efi/boot/grub.cfg using a text editor like nano, rather than the commands below, especially if you wish to position the entry in the grub menu

```
sudo cp /mnt/realboot/efi/boot/grub.cfg /mnt/realboot/efi/boot/grub.cfg.bak
sudo cat /mnt/realimages/My_New_OS_pi4/My_New_OS_pi4_grub.cfg | sudo tee -a /mnt/realboot/efi/boot/grub.cfg
```

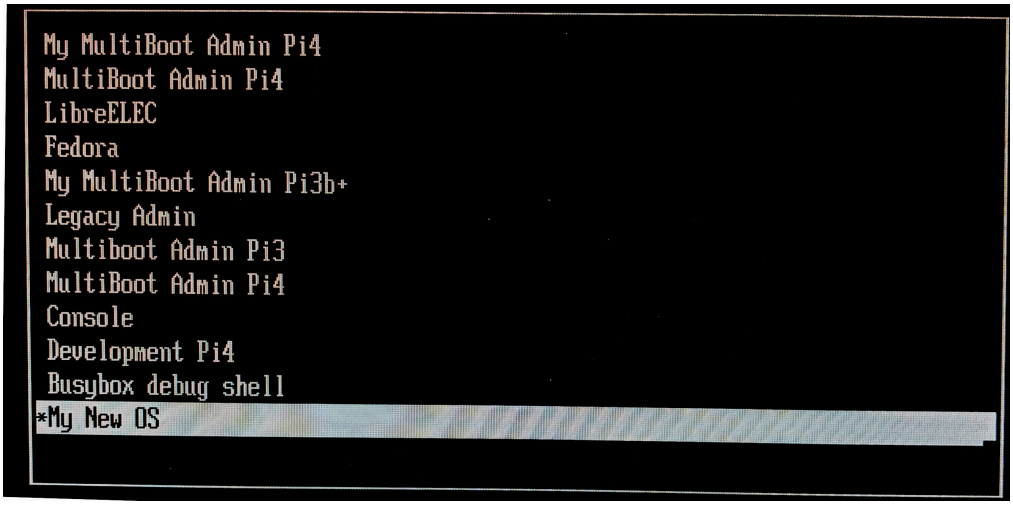
5) Unmount the partitions cleanly

With the Multiboot_Admin os use

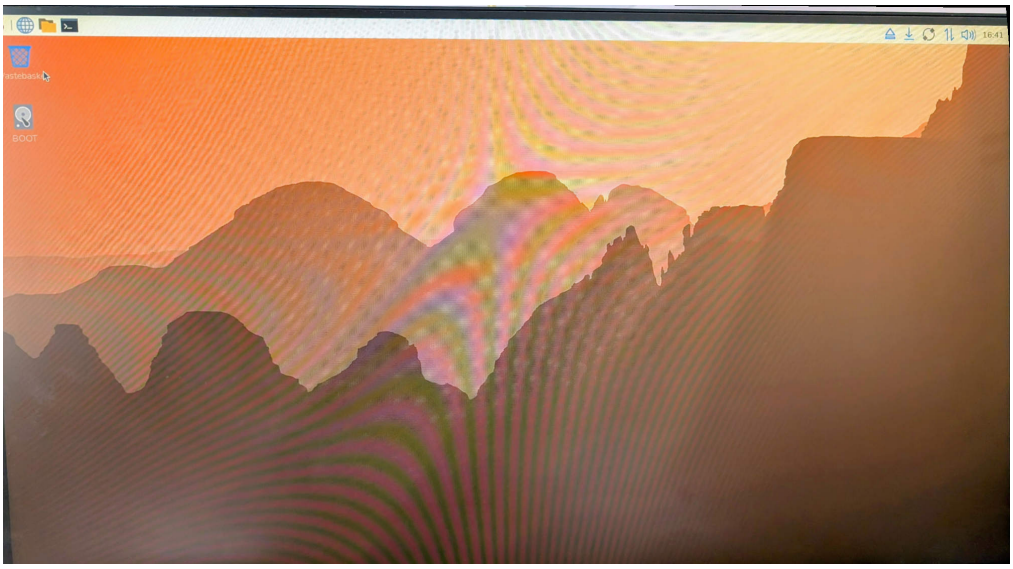
```
~/Pi_Multiboot/tools/unmountdrives.sh
```

6) Test

- Grub entry



- After Boot



- Note that
  - First boot of many OSes can take a long time
  - There will be some errors in the startup mainly due to upower (not sure what this is)
  - There is often an unexpected reboot after the very first boot. This may be the Raspbian resize functionality kicking in, I'm not sure.

Conclusion

All the 64 bit Raspian images and the 64 bit Ubuntu images that I've tested booted using this technique. All 4 processors are present and there is working sound but that's all that I've tested.

Footnote

Using some slight variations I've also managed to boot :-

- **LibreELECT** (v21.3 and only for the pi4 currently, due to lack of pi5 hardware and LibreELECT for the pi3 being 32 bit)
    - Use the same process as above to create the image
    - This then requires a small overlay initramfs (to include a custom "platform_init" file that loop mounts the LibreELECT image and also a custom Grub entry. LibreELECT files here
    - Unzip the archive into the same directory as that of the LibreELECT img file
    - platform_init is included for info only. It isn't needed
    - If the version of the LibreELECT image changes on the raspi-imager you will have to update the grub entry to reference the new boot=UUID and disk=UUID entries in the grub "rootopts" variable.
- **Fedora** (Fedora-Workstation-Disk-43-1.6.aarch64.raw.xz)
    - [From the official site] (https://fedoraproject.org/workstation/download/)
    - This also needs a specific grub entry and initramfs overlay.
    - The build process is much more manual. I'll provide details if anyone is interested
- I'm looking to see if I can do anything with **Lineage** but it would need quite a lot of detective work